

Centre Number	Candidate Number	Name
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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

PHYSICS

9702/06

Paper 6

May/June 2005

45 minutes

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen in the spaces provided on the Question Paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** of the questions in any **two** options.
The number of marks is given in brackets [] at the end of each question or part question.
You may lose marks if you do not show your working or if you do not use appropriate units.

If you have been given a label, look at the details. If any details are incorrect or missing, please fill in your correct details in the space given at the top of this page.

Stick your personal label here, if provided.

For Examiner's Use	
A	
F	
M	
P	
T	
Total	

This document consists of **21** printed pages and **3** blank pages.



Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2} QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = Ar\eta v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2 \rho v^2$$

Answer **all** of the questions in any **two** of the Options.

Answer the questions in the spaces provided on the Question Paper.

The Options are as follows.

Option A	Astrophysics and Cosmology	questions 1, 2 and 3
Option F	The Physics of Fluids	questions 4, 5 and 6
Option M	Medical Physics	questions 7, 8 and 9
Option P	Environmental Physics	questions 10, 11 and 12
Option T	Telecommunications	questions 13 and 14

Option A

Astrophysics and Cosmology

- 1 Fig. 1.1 illustrates the Milky Way galaxy.

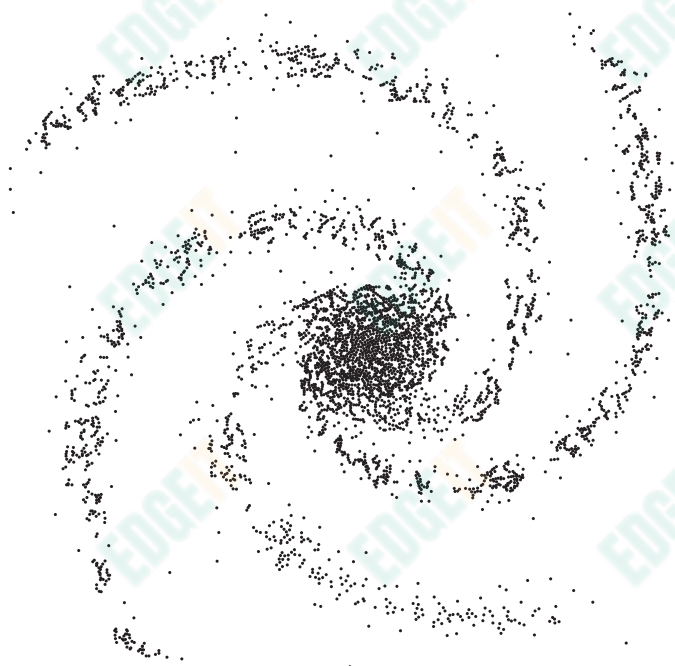


Fig. 1.1

- (a) On Fig. 1.1, mark the approximate position of the Solar System.

[1]

(b) State approximate values, in light-years, for

(i) the diameter of the Milky Way galaxy,

diameter = light-years [1]

(ii) the average separation of stars in one arm of the galaxy.

separation = light-years [1]

(c) Hubble was able to map out the Universe to a distance of approximately 500 million light-years.

State the number of galaxies that were estimated to exist within this distance.

number = [1]

2 Fig. 2.1 shows the variation with age of the temperature of the Universe.

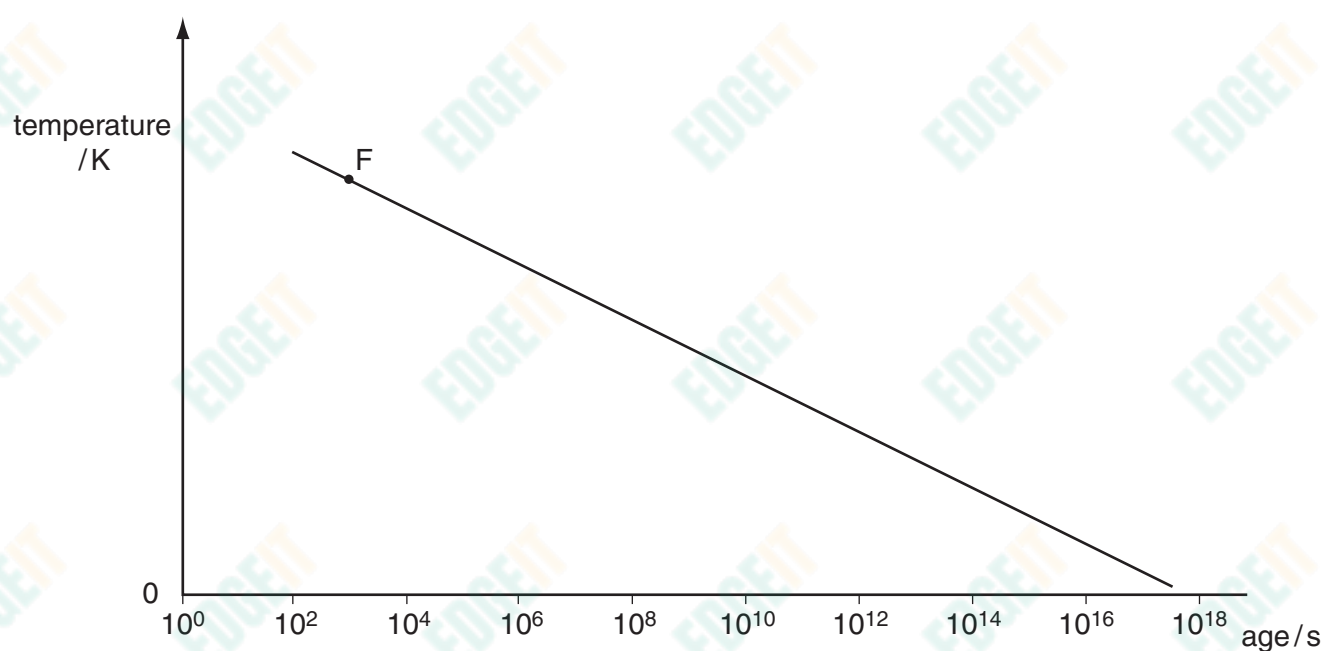


Fig. 2.1

- (a) Point F on the graph represents the point at which light elements were formed. Suggest the maximum temperature at which this formation occurred.

temperature = K [1]

- (b) Mark the point on the line at which galaxies began to form. [1]

- (c) Suggest why, in the Universe, there is an excess of matter over antimatter.

.....

.....

.....

.....

.....[4]

- 3 It has been reported recently that some galaxies have been discovered that are estimated to be 12600 million light-years from Earth.

- (a) (i) The Hubble constant has been estimated to be $60 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Use this value to calculate the approximate age of the Universe. ($1 \text{ pc} = 3.1 \times 10^{16} \text{ m}$)

age = years [5]

- (ii) Hence calculate the fraction of the age of the Universe for which the light from these galaxies has been travelling to Earth.

fraction = [1]

- (iii) Suggest why the discovery of these galaxies may have significance for an understanding of the development of the Universe.

.....

 [2]

- (b) Explain why there is a limit to the extent of the observable Universe.

.....

 [2]

Option F

The Physics of Fluids

- 4 A rectangular iceberg floats in seawater of density 1030 kg m^{-3} , as illustrated in Fig. 4.1.

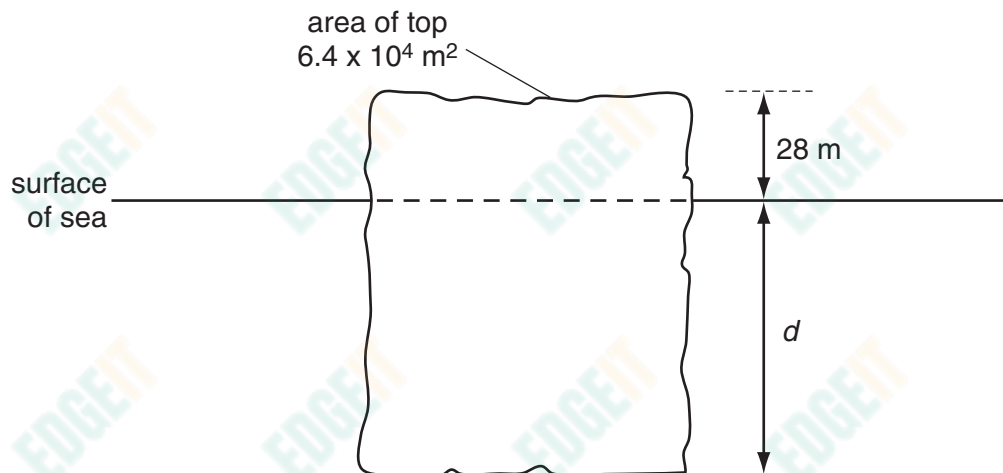


Fig. 4.1

The iceberg floats with its top, area $6.4 \times 10^4 \text{ m}^2$, 28 m above the surface of the sea. The density of ice is 920 kg m^{-3} .

- (a) State what provides the upthrust on an object when it is immersed in a fluid.

.....
[1]

- (b) The bottom of the iceberg is at a depth d , measured in metres. Give expressions, in terms of d , for

- (i) the mass of the iceberg,

.....

[2]

- (ii) the mass of seawater displaced by the iceberg.

.....
[1]

(c) Using your answers in (b), determine

(i) the depth d ,

$d = \dots\dots\dots$ m [2]

(ii) the fraction of the iceberg that is below the water surface.

fraction = $\dots\dots\dots$ [1]

- 5 (a) Explain what is meant by a *viscous* liquid.

.....

.....

.....[2]

- (b) Suggest why, for viscous flow of a fluid through a pipe, the flow rate is expressed as a volume flow rate, measured in $\text{m}^3 \text{s}^{-1}$, rather than a linear speed, measured in m s^{-1} .

.....

.....

.....[2]

- (c) The volume flow rate V of a viscous liquid undergoing steady flow through a tube of length L and radius r is given by

$$V = \frac{\pi r^4 p}{8\eta L},$$

where η is the viscosity of the liquid and p is the pressure difference between the ends of the tube.

Water of density $1.0 \times 10^3 \text{ kg m}^{-3}$ flows out of a container through a tube of length 13 cm and internal diameter 1.8 mm, as illustrated in Fig. 5.1.

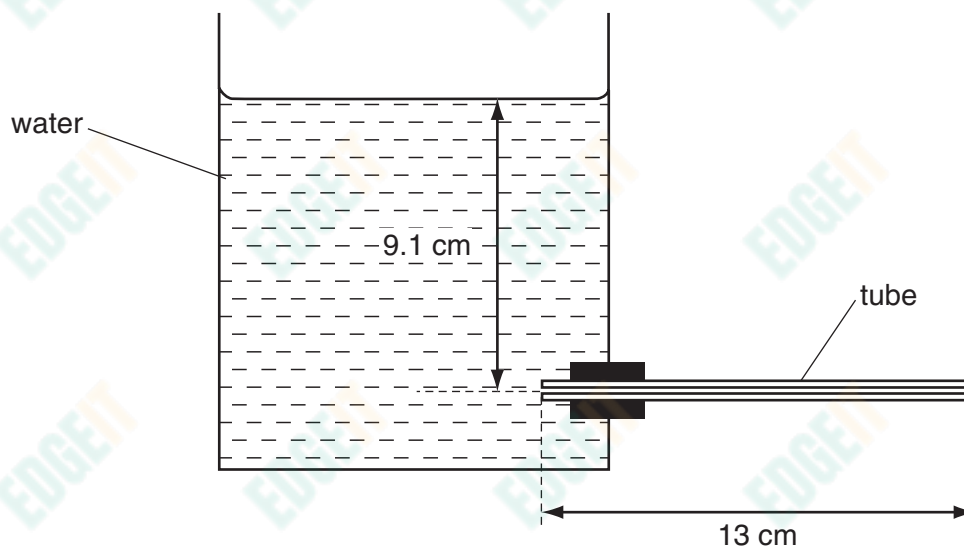


Fig. 5.1

When the axis of the tube is at a depth of 9.1 cm below the water surface, water runs out of the tube at a rate V of $1.5 \times 10^{-6} \text{ m}^3 \text{ s}^{-1}$.

- (i) Show that the pressure difference between the ends of the tube is approximately 890 Pa.

[2]

- (ii) Determine the viscosity of the water.

viscosity = N s m^{-2} [2]

6 Streamlines are a means by which fluid flow may be modelled.

(a) Explain

(i) what is meant by a *streamline*,

.....
.....[1]

(ii) why streamlines can never touch or cross.

.....
.....[1]

(b) State and explain the change in fluid flow when streamlines become closer together.

.....
.....
.....
.....[3]

Medical Physics

- [5]

1.
2.
-[2]

8 A person is accidentally exposed for a short time to ionising radiation.

(a) Explain

(i) what is meant by *absorbed dose*,

.....

.....

.....[2]

(ii) why the effect of the exposure depends on the type of radiation.

.....

.....

.....[2]

(b) Suggest why the effects of the exposure cannot be fully assessed within a few weeks of the exposure.

.....

.....

.....[2]

- 9 A person discovers that the power of the lens used to correct the defect of one of his eyes is +2.5 D.

(a) For this corrective lens,

(i) name the type of lens,

.....[1]

(ii) calculate the focal length in centimetres.

focal length = cm [1]

(b) (i) Name the defect of this eye.

.....[1]

(ii) Make calculations to estimate the distance of the far point and the distance of the near point from this unaided eye.

distance of far point =
distance of near point =
[4]

Option P

Environmental Physics

- 10 (a) Explain why the Earth's resources of fossil fuels are different from the total reserves.

.....

.....

.....

.....[3]

- (b) The formation of fossil fuels is a continuous process. Suggest why the reserves of these fuels are considered to be finite.

.....

.....

.....[2]

11 (a) Complete Fig. 11.1 to illustrate the action of a four-stroke petrol engine.

[4]

stroke	induction	compression	power	
inlet valve	open			closed
outlet valve				open

Fig. 11.1

(b) Modern four-stroke petrol engines frequently consist of four cylinders, each fitted with two inlet valves and two exhaust valves. Suggest, with one reason (in each case), why

(i) there are four cylinders,

.....

[2]

(ii) there are more than two valves in each cylinder.

.....

[2]

12 (a) (i) Explain what is meant by *pollution*.

.....
.....
.....[2]

(ii) State one example of pollution that is

1. man-made,

.....

2. natural.

.....[2]

(b) Suggest why large areas of forest are important to maintain the Earth's atmosphere and the environment.

.....
.....
.....
.....[3]

Option T

Telecommunications

13 An analogue signal is to be transmitted in digital form.

- (a)** Outline briefly the principles involved in the analogue-to-digital conversion of the signal and its transmission.

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

- (b)** Suggest two advantages of the transmission of data in digital form.

1.

.....

2.

.....[2]

- 14** It is proposed to install an optic fibre link between London (England) and Sydney (Australia). The optic fibre, length 18000 km, has an attenuation of 1.7 dB km^{-1} . The noise power in the fibre is $6.0 \times 10^{-19} \text{ W}$. The refractive index of the core of the fibre is 1.5.

(a) Explain what is meant by

(i) *attenuation*,

.....
[1]

(ii) *noise*.

.....
[1]

(b) The signal-to-noise ratio must not fall below 25 dB.

(i) Show that the smallest effective signal power that can be detected in the fibre is $1.9 \times 10^{-16} \text{ W}$.

[2]

(ii) Calculate the maximum uninterrupted length of fibre through which a signal can be transmitted for an input signal of power 7.0 mW.

length = km [3]

- (c) With reference to your answer in (b)(ii), suggest what can be done to enable the signal to be received in Sydney.

.....
.....[1]

- (d) An alternative means by which the signal could be sent is by using a geostationary satellite. The orbit of such a satellite is 3.6×10^4 km above the Earth's surface.

- (i) Explain what is meant by a *geostationary orbit*.

.....
.....
.....[2]

- (ii) By considering the times of transmission by optic fibre and by satellite, explain briefly one advantage of the fibre when compared with the satellite.

.....
.....[3]

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